

The Number That Refused to Move

A case for treating fractions as permanent residents of the number line,
rather than instructions to reach for a calculator.

Written for teachers, written for learners, written for anyone
who has ever pressed $\div$ and watch the fruits of human ingenuity wither.

§ 1 EVERY TIME YOU PRESS $\div$, SOMETHING DIES

There is a small violence that happens in classrooms every day, so routine that no one notices it. A child sees $6/9$ and, trained by primary instruction, reflexes for the calculator. Button pressed a number 0.6666667 appears on the screen. The child writes it down, and the lesson continues.

What died? The relationship. The fact that 6 and 9 share a factor of 3. The fact that this is the same point on the number line as $\frac{2}{3}$, with a cleaner name.

The fact that this fraction can be added to $\frac{1}{3}$ to make *exactly* 1 — 0.9999999 or 1.0000001 nowhere in sight. All of that structure, all of that legibility, incinerated the moment the divide button was pressed.

THE VERB PATH

0.6666667

Structure destroyed.

Rounding error installed.

THE NOUN PATH

 $\frac{2}{3}$

Structure preserved.

Exact mathematical reality.

We have spent decades optimising the wrong thing training children to turn the handle, and then the spreadsheet arrived. The handle-turning jobs never quite evaporated. They really should have. The calculator had long since put paid to long division and slide rulers. Now AI is mopping up these trifling algorithmic task. Rightly so. And still, in classrooms across the country, children are pressing $\div$ and destroying the very structure they should be learning to see.

§ 2 A FRACTION IS A NOUN, NOT A VERB

Here is the central claim: **the fraction bar is a name, not an instruction.**

When a child reads $6/9$ and thinks “six divided by nine,” they are reading a noun as a verb and so treating a permanent address on the number line as a mere command to perform a calculation. This is the root of many a number line and algebraic trouble that runs deep through secondary school.

A fraction is a number in its own right. It does not demand division.

Consider how mathematicians and engineers actually think about fractions. The probability of a fair coin landing heads is $\frac{1}{2}$. The difference between writing $\frac{1}{2}$ and writing 0.5 is categorically significant: in a cryptographic proof, or a tolerance calculation, or a probability argument, *exactly* one half and *approximately* 0.5 or indeed 0.50 are different claims.

The Egyptians understood this four thousand years ago. Their entire mathematical system was built on unit fractions — numbers of the form $\frac{1}{n}$. They did not press divide. They reasoned about relationships: which slices fit into which wholes, which denominators could be combined, how one quantity compared to another, all without reducing everything to a decimal smear. The *Rhind Mathematical*

Papyrus, dated to around 1650 BCE, contains sophisticated fraction arithmetic that a modern calculator-reliant student would find opaque, precisely because it requires reading fractions as objects rather than operations.

In the age of AI, this distinction sharpens rather than blurs. Every large language model, every optimiser, every automated system is by nature a *verb machine*. It processes, transforms, outputs. It turns handles with extraordinary speed and competence. What it cannot do is what only a mind trained in genuine mathematical reasoning can do: that is to *see structure*. And structure lives in the fraction, not the decimal.

§3 MATHEMATICS IS MOSTLY ABOUT INEQUALITY

Schools teach equality as the destination of all mathematical activity. Solve for x . Find the answer. Get the number. This is the exam-shaped version of mathematics, optimised for marking, hostile to genuine reasoning.

§4. INEQUALITY AND EGYPTIAN STRUCTURE

Real mathematics, the kind that builds bridges, designs circuits, and models climate systems¹, operates through comparison. The central questions take the form

larger than, smaller than, bounded by, close to.

Fractions support these comparisons directly. Decimal representations tend to conceal the internal structure that drives them.

EGYPTIAN COMPARISON WITHOUT A CALCULATOR

$$\frac{8}{11} \quad \text{vs} \quad \frac{10}{17} \quad \text{vs} \quad \frac{8}{12}$$

Each fraction can be written as a deviation from $\frac{1}{2}$ using unit fractions:

$$\frac{8}{11} = \frac{1}{2} + \frac{5}{22}, \quad \frac{10}{17} = \frac{1}{2} + \frac{3}{34}, \quad \frac{8}{12} = \frac{2}{3} = \frac{1}{2} + \frac{1}{6}.$$

¹Inequality also dominates in analytic number theory and combinatorics, where bounding arguments and extremal behaviour are central.

Now express each remainder in Egyptian form:

$$\frac{5}{22} = \frac{1}{5} + \frac{1}{110}, \quad \frac{3}{34} = \frac{1}{12} + \frac{1}{204}, \quad \frac{1}{6} \text{ already unit.}$$

Comparison reduces to comparing leading unit fractions:

$$\frac{1}{5} > \frac{1}{6} > \frac{1}{12}.$$

Hence the ordering follows immediately:

$$\frac{10}{17} < \frac{8}{12} < \frac{8}{11}.$$

The method generalises. For any fraction

$$\frac{a}{b},$$

one can locate it relative to a benchmark such as $\frac{1}{2}$, $\frac{1}{3}$, or $\frac{2}{3}$, and then decompose the residual into unit fractions. The leading unit fraction dominates the comparison, because

$$\frac{1}{n} > \frac{1}{m} \iff n < m.$$

The ordering of fractions becomes an ordering of denominators.

The greedy algorithm formalises this process. At each step, select the largest unit fraction not exceeding the remaining quantity:

$$\frac{a}{b} = \frac{1}{\lceil b/a \rceil} + \text{remainder}.$$

Iteration produces a decomposition

$$\frac{a}{b} = \frac{1}{n_1} + \frac{1}{n_2} + \cdots + \frac{1}{n_k}, \quad n_1 < n_2 < \cdots < n_k.$$

The first denominator n_1 sets the scale. Subsequent terms refine the position.

The identity

$$\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}$$

exhibits how unit fractions split into finer components. Repeated application generates increasingly precise decompositions while preserving exact equality.

Prime denominators impose longer decompositions under the greedy algorithm. Composite denominators often admit shorter representations due to internal factor structure. This reflects the arithmetic of the denominator directly.

The governing principle is simple:

Smaller denominator, larger unit. Larger denominator, finer correction.

Fractions written in this form display their magnitude through their structure. The comparison becomes visible without numerical approximation.

We train children for equality. Real mathematics lives in the inequalities. Fractions are the native language of those inequalities. Decimals are a translation, and something is always lost in translation.

§ 4 THE LEDGER: A RECORD OF HOW THINKING CHANGES

There is much talk of a ledger being the answer to a student providing proof of providence in their learning. Here is a fraction ledger. Two columns: what the fraction is as a verb, what it is as a noun. The pedagogical claim is this: learning to write in both columns, and to notice when one is tempted to collapse the noun into the verb, is itself an act of mathematical formation. It is the habit that separates a mathematician from a calculator.

#	THE VERB IMPULSE	THE NOUN READING
01	Six divided by nine. I wanted to press $\div$. Result would be $0.666\dots$	$\frac{6}{9}$ is the number two-thirds, living at the same address as $\frac{4}{6}$, $\frac{10}{15}$. The simplest name: $\frac{2}{3}$.
02	Three-quarters plus a fifth. I added the bottom numbers and got $\frac{2}{9}$.	The denominator is a slice size. Common architecture needed: twentieths. The answer is $\frac{9}{20}$.
03	$\frac{3x}{18} = \frac{1}{12}$. I converted both sides to decimals. Got $x \approx 0.49997$.	Cancel the 3s first: $\frac{x}{6} = \frac{1}{12}$. Multiply by 6: $x = \frac{6}{12} = \frac{1}{2}$. Proven, not guessed.
04	The probability of heads is 0.5. The decimal felt more precise.	The probability of heads is $\frac{1}{2}$. That is the exact structure of the event. The decimal hides this.
05	$\frac{3}{7}$ seemed hard so I reached for a calculator: $0.428571\dots$	$\frac{3}{7} = \frac{1}{3} + \frac{1}{11} + \frac{1}{231}$. The greedy algorithm finds the structure. 7 is a reptend prime: its fractions cycle. The decimal hid the cycle entirely.

Each row is a course correction. Each entry is a small act of intellectual honesty: I was going to do this, but I noticed that doing so would destroy something true. This noticing is what we are trying to cultivate. No AI will cultivate it for you. No calculator will teach it. The habit must be built by hand, entry by entry, the way any real ledger is kept.

The whole purpose of the ledger is to record the moment you noticed you were about to stop thinking.

§ 5 THE PROTOCOL: RESIST, RETAIN, READ

A four-step procedure, learnable by any child, teachable by any teacher. It requires no special equipment — only a pencil and a refusal to press $\div$ before looking at the structure.

- 1. Resist.** Before touching a calculator or performing any division, look at the fraction as a name. Say it aloud as a noun: “the number three-sevenths.” Feel the difference from “three divided by seven.”
- 2. Simplify by cancelling.** Find the GCD of numerator and denominator. Cancel the shared factor. The number has changed its name, not its address.

$\frac{6}{9}$ was always $\frac{2}{3}$. The calculation finds the cleaner label.

3. **Retain.** Keep everything in fraction form for as long as possible. Do the algebra symbolically. Cancel across fractions before multiplying. Fractions multiply cleanly; decimals compound error at every step.
4. **Read the exact form.** When you arrive at your answer, read it as a noun. $x = \frac{1}{2}$, not 0.5, not approximately 0.49997. If a decimal is required for practical purposes, convert at the final step, and only then.

This protocol is not anti-technology. It is pro-structure. The calculator remains a powerful tool for the final step and for checking results. The resistance is to *premature collapse* — the habit of converting a noun into a number before understanding what the noun is saying.

In an age when AI will perform every calculation faster and more accurately than any human, the value of the human mathematician lies precisely here: in the capacity to hold a structure in mind, to see what the fractions are telling you before computing, to know that the answer is *exactly* one half before arithmetic confirms it, because the structure made it inevitable. That capacity cannot be automated. It can only be trained, and it is trained by treating fractions as nouns.

Little mathematicians are handle-readers, not handle-turners. The ledger is the practice. The fraction is the text. The lesson: **look before you calculate, and sometimes looking is enough.**
